Osmerus mordax (Rainbow Smelt) Fishes Native Transplant

Collection Info
Point Map
Species Profile
Animated Map

Wayne Nelson-Stastny - South Dakota Game, Fish, and Parks ©

Osmerus mordax (Mitchill, 1814)

Common name: Rainbow Smelt

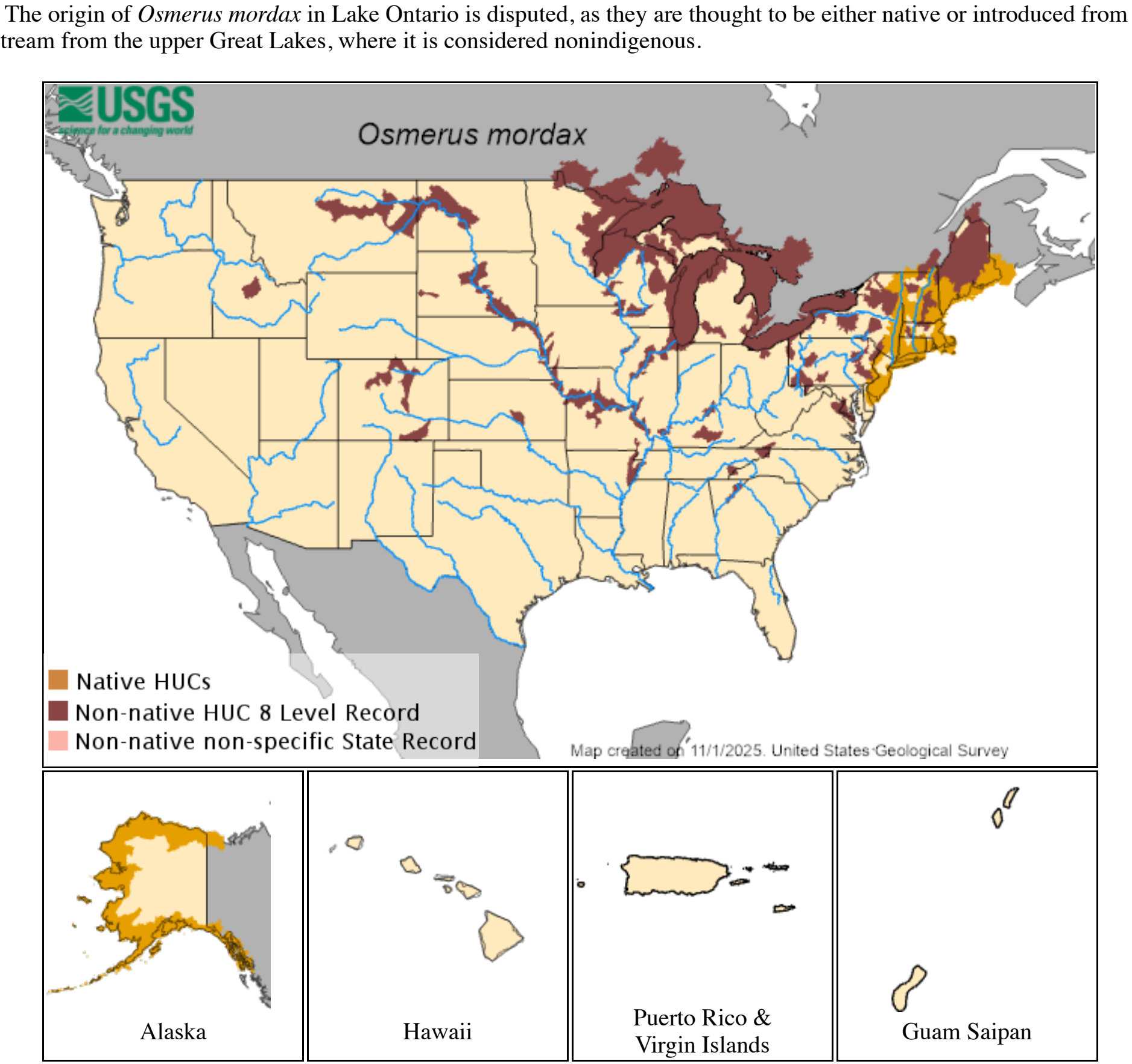
Synonyms and Other Names: *Atherina mordax* (Mitchill, 1814), *Osmerus spectrum* (Cope, 1870), American Smelt, Atlantic Smelt, Freshwater Smelt, Leefish.

Taxonomy: available through ITIS

Identification: Rainbow Smelt have a silvery, pale green back and a white underside. The sides of the fish are iridescent purple, blue, and pink with a bright silvery stripe. They have less than 75 scales on the lateral line, which are easily removed. Rainbow Smelt have three characteristic fins, a dorsal in the middle of the back, an adipose, and a deeply forked rear tail fin. The snout is pointed with well developed, canine-like teeth on the tongue and roof of the mouth. Rainbow Smelt resemble the minnow family (*Cyprinidae* spp.), Lake Whitefish (*Coregonus clupeaformis*) and Cisco (*C. artedi*), however, minnows lack an adipose fin and Lake Whitefish and Cisco have deeper bodies than Rainbow Smelt but no teeth on the roof of their mouths or tongues. See Scott and Crossman (1973); Becker (1983); Smith (1985); Page and Burr (1991).

Size: Adult: 100 mm to 350 mm. Egg: 0.90 mm to 1.3 mm (Scott and Crossman 1973)

Native Range: Atlantic drainages from Lake Melville, Newfoundland, to Delaware River, and Pennsylvania; Arctic and Pacific drainages from Bathurst Inlet, Northwest Territories, to Vancouver Island, British Columbia. Also, Pacific drainages of Asia (Page and Burr 1991). The origin of *Osmerus mordax* in Lake Ontario is disputed, as they are thought to be either native or introduced from the Atlantic through the Erie Canal (Mills et al. 1993). Another alternative is that *O. mordax* migrated downstream from the upper Great Lakes, where it is considered nonindigenous.


USGS science for a changing world
Osmerus mordax
Native HUCs
Non-native HUC 8 Level Record
Non-native non-specific State Record
Map created on 11/1/2025. United States Geological Survey
Alaska
Hawaii
Puerto Rico & Virgin Islands
Guam Saipan

Native range data for this species provided in part by NatureServe
Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Osmerus mordax* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AR	1978	1981	3	Lower Mississippi-Greenville; Lower Mississippi-Memphis; Lower St. Francis
CO	1973	2019	6	Cache La Poudre; Colorado Headwaters; Fountain; Middle South Platte-Cherry Creek; Purgatoire; South Platte
CT	1930	1940	3	New England Region; Outlet Connecticut River; Shetucket River
GA	1971	1971	1	Upper Chattahoochee
ID	1930	1962	2	Upper Salmon; Upper Snake
IL	1923	2016	11	Cahokia-Joachim; Chicago; Copperas-Duck; Des Plaines; Lake Michigan; Little Calumet-Galien; Lower Illinois-Lake Chautauqua; Lower Illinois-Senachwine Lake; Lower Ohio-Bay; Upper Illinois; Upper Mississippi-Cape Girardeau
IN	1923	2010	3	Lake Michigan; Little Calumet-Galien; Silver-Little Kentucky
IA	1975	1987	3	Big Papillion-Mosquito; Blackbird-Soldier; Keg-Weeping Water
KS	1978	2011	3	Independence-Sugar; Little Arkansas; Tarkio-Wolf
KY	1978	1986	2	Lower Mississippi-Memphis; Upper Green
LA	1979	1979	1	Lower Mississippi-Baton Rouge
ME	1965	2020	21	Allagash River; Aroostook River; Dead River; East Branch Penobscot River; Fish River; Headwaters Saint John River; Lower Androscoggin River; Lower Kennebec River; Maine Coastal; Mattawamkeag River; Meduxnekeag River; New England; New England Region; Penobscot River; Piscataqua-Salmon Falls; Piscataquis River; Presumpscot; Saint Francis River-Saint John River; St. George-Sheepscot; Upper Androscoggin River; Upper Kennebec River; West Branch Penobscot River
MD	1875	1979	3	Mid Atlantic Region; Middle Potomac-Anacostia-Occoquan; Youghiogheny
MA	1833	2005	5	Ashuelot River-Connecticut River; Charles; Nashua River; New England Region; Westfield River
MI	1906	2024	17	Betsie-Platte; Betsy-Chocolay; Boardman-Charlevoix; Cheboygan; Dead-Kelsey; Detroit; Escanaba; Great Lakes Region; Kalamazoo; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Lake Superior; St. Clair; St. Marys; Upper Wisconsin
MN	1945	2025	16	Baptism-Brule; Beaver-Lester; Buffalo-Whitewater; Coon-Yellow; Kettle; Lake of the Woods; Lake Superior; Lower St. Croix; Prairie-Willow; Rainy; Rainy Headwaters; Rainy Lake; Snake; St. Croix; St. Louis; Vermilion
MO	1978	2011	11	Blackwater; Cahokia-Joachim; Independence-Sugar; Lake of the Ozarks; Little Chariton; Lower Mississippi-Memphis; Lower Missouri; Lower Missouri-Crooked; Lower Missouri-Moreau; Tarkio-Wolf; Upper Mississippi-Cape Girardeau
MT	1979	2011	6	Bullwhacker-Dog; Charlie-Little Muddy; Fort Peck Reservoir; Lower Yellowstone; Milk; Prairie Elk-Wolf
NE	1975	2007	3	Big Papillion-Mosquito; Keg-Weeping Water; Lewis and Clark Lake
NH	1929	1994	14	Ashuelot River-Connecticut River; Black River-Connecticut River; Contoocook River; Headwaters Connecticut River; Merrimack River; Millers River; New England; Pemigewasset River; Piscataqua-Salmon Falls; Saco River; Upper Androscoggin River; Waits River-Connecticut River; West River-Connecticut River; Winnipesaukee River
NY	1917	2019	15	Black; Lake Erie; Lake Ontario; Long Island; Middle Delaware-Mongaup-Brodhead; Niagara River; Northern Long Island; Oak Orchard-Twelvemile; Oswegatchie; Raquette; Sacandaga; Seneca; Upper Hudson; Upper Susquehanna; Upper Susquehanna
NC	1991	1991	1	Upper Tennessee
ND	1971	2022	2	Lake Sakakawea; Painted Woods-Square Butte
OH	1935	2017	2	Black-Rocky; Lake Erie
PA	1862	2017	9	Allegheny; Chautauqua-Conneaut; Lake Erie; Lower Monongahela; Middle Allegheny-Redbank; Raystown; Schuylkill; Shenango; Upper Susquehanna-Lackawanna
SD	1974	2005	4	Fort Randall Reservoir; Lewis and Clark Lake; Lower Lake Oahe; Rapid
TN	1956	1980	4	Lower Little Tennessee; Lower Mississippi-Memphis; South Fork Holston; Watauga, North Carolina, Tennessee
VT	1991	1994	3	St. Francois River; Waits River-Connecticut River; Winooski River
VA	1891	1994	3	Lower Potomac; Middle Potomac-Anacostia-Occoquan; Potomac
WI	1887	2020	23	Bad-Montreal; Beartrap-Nemadji; Brule; Coon-Yellow; Door-Kewaunee; Flambeau; Lake Michigan; Lake Superior; Lower Fox; Lower St. Croix; Lower Wisconsin; Manitowoc-Sheboygan; Menominee; Milwaukee; Namekagon; Ontonagon; Pike-Root; Red Cedar; St. Louis; Upper Chippewa; Upper St. Croix; Upper Wisconsin; Wolf

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: Rainbow Smelt can either be anadromous or landlocked in freshwater. They inhabit a large portion of the water column that varies throughout their life. Young of the year Rainbow Smelt typically are found along shorelines in warmer waters 2 to 6 meters deep. Yearlings are found in cooler water within the pelagic zones of lakes (Rooney and Paterson 2009). Adults are reported to undergo diel vertical migration and have been found at depths in Lake Superior and Lake Michigan between 18 and 91 meters depending on the season (Dryer 1966; Wells 1968; Becker 1983). Rainbow Smelt were found in high abundance in river plumes from major tributaries to Lake Michigan. This abundance may be a symptom of the river plumes' high productivity and the spawning habits of the species (Smith and Simpkins 2018).

Spawning of Rainbow Smelt occurs after ice-out in early spring once water temperatures reach ~4.4°C with a peak at 10°C (Becker, 1983). In the Great Lakes, spawning occurs in tributary streams and occasionally on the shoreline. Eggs of the Rainbow Smelt are adhesive, negatively buoyant, and adhere to substrate until hatching 2 to 3 weeks later (Scott and Crossman 1973; Becker 1983). Age of maturation ranges between 1 to 4 years amongst populations (McKenzie 1958; Murawski and Cole 1978) and life expectancy can reach 8 years (Bailey 1964; Kirn and Labar 1996). They are sexually dimorphic, with females growing larger and living longer than males (McKenzie 1958; Bailey 1964).

Rainbow Smelt can survive in a broad range of environmental conditions. They inhabit temperatures from -1.8°C to around 20°C (Nellbring 1989; Raymond 1995), and can acclimate to extreme cold (< 5°C) by producing an antifreeze protein, glycerol, to facilitate faster-swimming speeds and increased muscle function (Coughlin et al. 2019). A laboratory study that assessed embryonic and larval Rainbow Smelt survival in response to various abiotic factors (dissolved oxygen, pH, salinity, nitrates, and phosphates) found that their survival was only significantly reduced when dissolved oxygen levels were less than 20% and the pH<5 (Fuda et al. 2007). However,a review of data from 8,842 Ontario lakes found that Rainbow Smelt were only caught in lakes with a pH>6 despite adequate opportunity to invade lakes with lower pH values (Evan and Loftus 1987).

Rainbow Smelt are zooplanktivorous at small sizes (<150 mm SL), consuming copepods and cladocerans (e.g., *Daphnia* and *Bosmina*); while at larger sizes (>150 mm SL) diet includes benthic crustaceans, and larval and small fishes (Sheppard et al. 2012). They are an important forage fish for many predators, including Atlantic Salmon (*Salmo salar*), Walleye (*Sander vitreus*), and Lake Trout (*Salvelinus namaycush*) (Gorman 2019).

Means of Introduction: The earliest known record is from 1912, when eggs were stocked in Crystal Lake, Michigan, which drains into Lake Michigan (Van Oosten 1937). Fish escaped into Lake Michigan and spread quickly throughout the Great Lakes and their tributaries (Creaser 1926; Gerking 1945; Hubbs and Lagler 1947; Nelson and Gerking 1968; Christie 1974; Eddy and Underhill 1974; Smith 1979; Morrow 1980; Phillips et al. 1982; Cooper 1983; Emery 1985). Early records documenting the smelt's range expansion in the Great Lakes include Lake Michigan, 1923 (Christie 1974; Emery 1985), Lake Erie, 1935 (Cooper 1983; Smith 1985), Lake Huron, 1925 (Christie 1974; Eddy and Underhill 1974), Lake Ontario, 1929 (Christie 1974; Smith 1985), and Lake Superior, 1923 (Emery 1985). The Lake Ontario population may be either native to this lake or migrated downstream, possibly through the Welland Canal (Emery 1985; Smith 1985). Another possibility is that the species was introduced from the Finger Lakes via the Seneca-Cayuga, Erie, and Oswego canals (Smith 1985).

Two means have been proposed to explain the introduction of Rainbow Smelt into the Missouri and Mississippi rivers. It may have spread from Lake Michigan via the Chicago sanitary canal to the Illinois River and then to the Mississippi and Missouri rivers (Burr and Mayden 1980). Alternatively, the species may have gained access to these rivers as a result of a stocking at Lake Sakakawea, North Dakota, in 1971 (Bouc 1987; Mayden et al. 1987; Holton 1990). The second explanation seems more plausible because of a lack of records from the Illinois River. Records of first occurrences in other areas include the Mississippi River, Illinois, and Kentucky, 1978; Mississippi River, Louisiana, 1979; Mississippi River, Tennessee, and Arkansas, 1980; Missouri River, Missouri, 1980; Missouri River, Kansas, 1982 (Mayden et al. 1987). Mayden et al (1987) provided a map of the species' distribution, dates of first observation in new areas, and possible introduction pathways. The species was originally introduced into Lake Sakakawea, North Dakota, as forage for Salmonids (Mayden et al. 1987).

Status: Introduced populations of this species have been very successful and the Rainbow Smelt is now established in the Great Lakes and in most rivers and lakes where introduced. This species has done so well in the Great Lakes that a commercial fishery targeting Rainbow Smelt has been operating there for many years (Smith 1985). It is the most abundant fish in some samples taken from the Mississippi River (Pflieger 1997). Nevertheless, no adults of the Rainbow Smelt have been found in either Missouri (Pflieger 1997) or Tennessee (Etnier and Starnes 1993). As such, Pflieger (1997) concluded that populations in Missouri are maintained by the continued escape of fish from upstream reservoirs on the Missouri River. As of 1987, only one specimen had been taken from the Ohio River (Mayden et al. 1987). It is considered extirpated in Georgia; the species has not been observed in that state since its release (Dahlberg and Scott 1971b).

Great Lakes
Widespread invasive with high impact. Reproducing and overwintering at self-sustaining levels have been recorded in all five Great Lakes.

Impact of Introduction: Dominant prey form for salmonids; contributed to extinction of Blue Pike; affect imperiled species (U.S.EPA 2008; Crossman 1991). In the Great Lakes, Rainbow Smelt compete with Lake Herring *Coregonus artedii* for food (Becker 1983). Christie (1974) supplied some evidence to support this, correlating lake herring decline with smelt increases in most of the lakes. Todd (1986) also reported that smelt may be partially responsible for the decline of whitefish *Coregonus* spp. in the Great Lakes. Havey (1973) reported increased growth of landlocked Atlantic Salmon following the introduction of smelt as a forage species in a lake in Maine. Hrabik et al. (1998) found evidence of competition for food between introduced Rainbow Smelt and native Yellow Perch *Perca flavescens* in Wisconsin lake habitats. Stedman and Argyle (1985) found the diet of *O. mordax* to include young-of-the-year fish. The species that were consumed by *O. mordax* depended upon the prey availability (Stedman and Argyle 1985). Stedman and Argyle (1985) discovered that in Lake Michigan in late October of 1982, *O. mordax* consumed Bloaters (*Coregonus hoyi*) and Alewives (*Alosa pseudoharegus*). The authors also noted that although *O. mordax* does not appear to have had an impact on Bloater populations in the last five years, future impact is possible. A study on nighttime consumption of *O. mordax* in Lake Ontario revealed their primary food source to be Slimy Sculpin (*Cottus cognatus*) and opossum shrimp (*Mysis relicta*) (Brandt and Madon 1986). Juvenile Lake Trout (*Salvelinus namaycush*) rely heavily on *C. cognatus*, competing directly with *O. mordax*, while adult *S. namaycush* consume *O. mordax* (Brandt and Madon 1986). The authors point out that *S. namaycush* may be a keystone predator in the relationship between *O. mordax* and *C. cognatus*. Rainbow Smelt contain high levels of thiaminase, which reduces absorption and assimilation of thiamine in predators such as salmonids can cause reduced body condition, swim performance, and other potenial impacts (Houde et al. 2015)

Remarks: This species is eaten by humans and used as bait for Salmonids and Walleye (Pflieger 1997). O'Brien et al. (2014) investigated ecological factors influencing recruitment of Rainbow Smelt in Lake Huron, and suggest that the primary drivers on recruitment were cannibalism by older smelt, availability of spawning habitat due to spring precipitation, and predation on adult smelt by Lake Trout. Feiner et al. (2015) examined recruitment dynamics of Rainbow Smelt in Lake Michigan and found that significant variation in stock productivity over time had a strong influence on recruitment.

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Other Resources:
Distribution in Illinois - ILNHS

Great Lakes Indian Fish and Wildlife Commission Maps

Fishes of Wisconsin (Becker)

Great Lakes Waterlife

FishBase Summary

Author: Fuller, P., E. Maynard, J. Larson, A. Fusaro, T.H. Makled, M. Neilson and A. Bartos

Revision Date: 9/25/2020

Peer Review Date: 8/13/2020

Citation Information:
Fuller, P., E. Maynard, J. Larson, A. Fusaro, T.H. Makled, M. Neilson and A. Bartos, 2025, *Osmerus mordax* (Mitchill, 1814): U.S. Geological Survey, Nonindigenous Aquatic Species Database, Gainesville, FL, https://nas.er.usgs.gov/queries/FactSheet.aspx?SpeciesID=796, Revision Date: 9/25/2020, Peer Review Date: 8/13/2020, Access Date: 12/3/2025

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